Life Below Water

An illustrated adventure in ocean discovery

An educational game that takes children from a sunny shoreline into the deep sea on a four-metre illustrated map, steering a robot ball and following a talking shell — learning about marine life, pollution and recycling on the way.

DISCIPLINE
Graphic & Illustration

YEAR
2020–2021

ROLE
Visual concept, map & lesson boards, marine illustration, storytelling, activity sheets

FORMAT
4 × 1 m illustrated map · Sphero robot · children aged 9



02 — BEFORE YOU BEGIN

What you'll find here

Life Below Water took more than eight months: a teaching programme about oceans, pollution and recycling, turned into a four-metre illustrated map that children explore with a robot ball.

This file follows the work in that order — the brief and the game first, then the map stage by stage, then the characters, the colour system and the process behind them.

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HOW TO READ

Three conventions run through the boards.



Each stage keeps its colour throughout the file, from sunny yellow to deep violet-blue.

5 min Minutes per block come from the project's own lesson plan.



Line drawings are the project's storyboard; colour pieces are final artwork.

03 — THE BRIEF

A whole programme, for nine-year-olds

The starting point was a substantial teaching programme: why oceans matter, what lives in them, what pollutes them and how recycling helps. The aim was to let children face marine pollution and the idea of sustainability as something they experience, not a lecture they sit through.

THE DESIGN CHALLENGE

Turn the programme into a journey a child can understand — and wants to explore.

AUDIENCE

Children around age 9 — old enough to start thinking about the world, its problems and its future.

LEARNING GOALS

01 Get to know the ocean and why it matters

02 Recognise different types of garbage

03 Learn how to clean the environment and recycle

04 Understand what sustainability means

WHAT THEY LEARN

From the lesson content

70%
of the planet is ocean

97%
of Earth's water is in it

3 bn
people depend on marine ecosystems

80%+
of the ocean is unexplored

04 — IN THE BOX

Unbox, scan, set off

The game arrives as a box. Children unpack it, meet the objects one by one, and the story starts the moment they scan the shell.

01 The map

4 × 1 m, laid on the floor: four connected stages from beach to deep sea.

02 Sphero

A robotic ball the child steers across the map from a phone app.

03 The shell

A real shell with a QR code. Scanned, it speaks — and asks for help.

04 The die

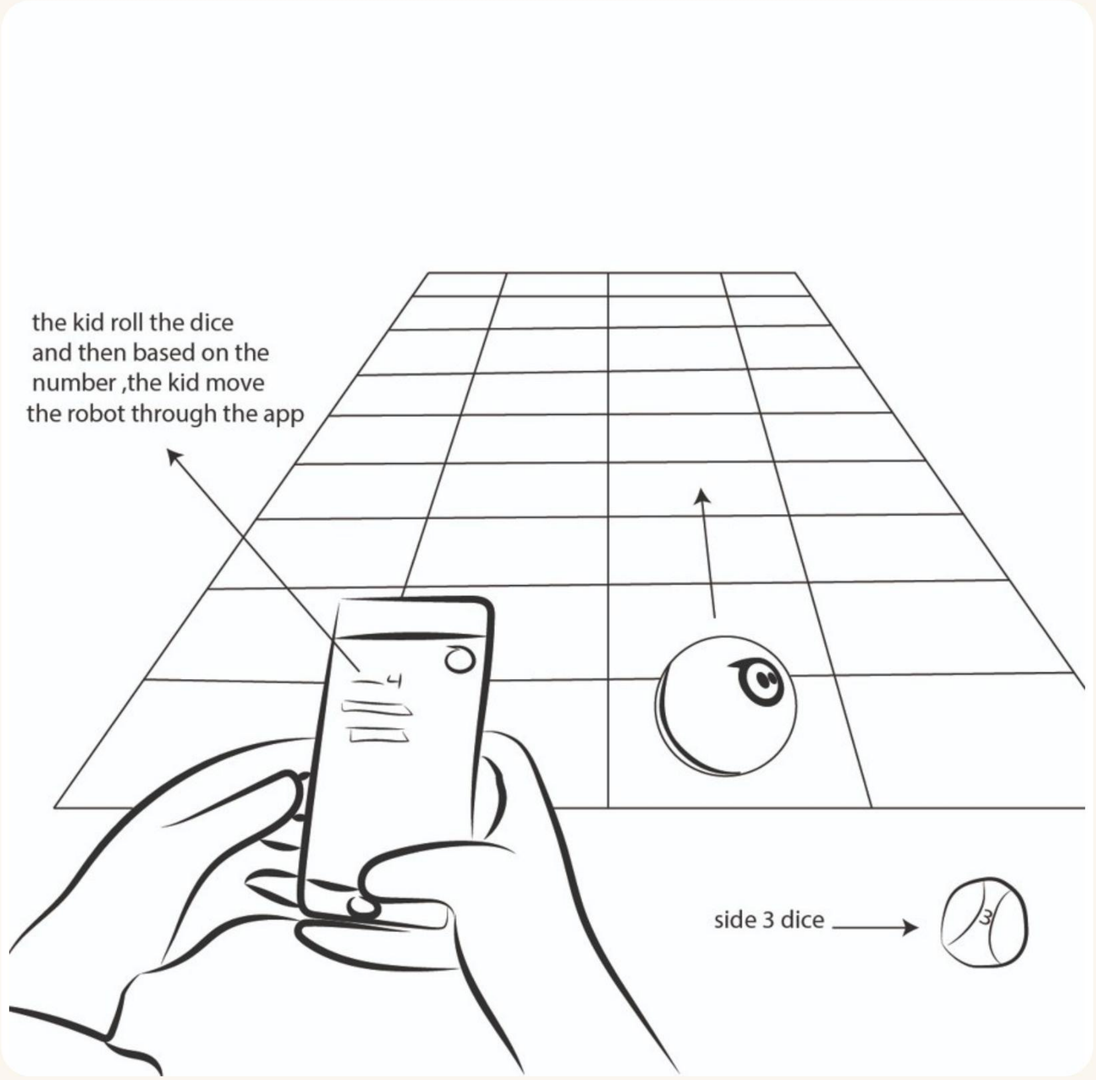
A three-sided die decides how many blocks to move.

05 The phone

Scans each block's QR code to open a video, a story or a question.

06 The guidebook

Read aloud by the teacher, who plays the game's helper.



Storyboard · roll the die, then drive Sphero that many blocks

The map is a grid of blocks — each one an activity

05 — HOW A TURN WORKS

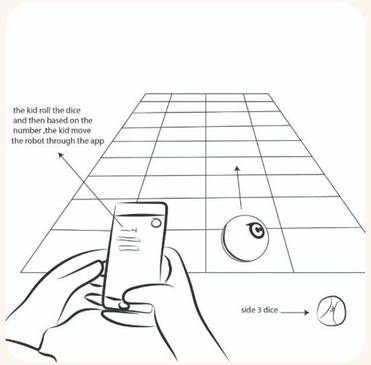
One loop, block by block

The interaction had to work for a nine-year-old without instructions on screen: a physical roll, a robot to steer, a picture to scan. The phone only ever does one thing at a time.



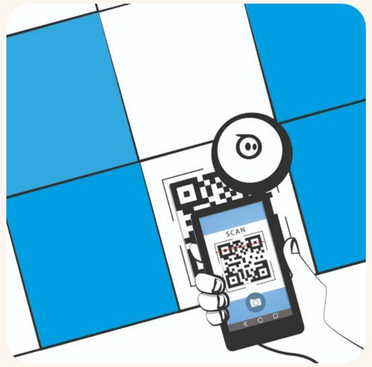
01 The shell calls

Scanning the real shell plays its message: the creatures below need help.



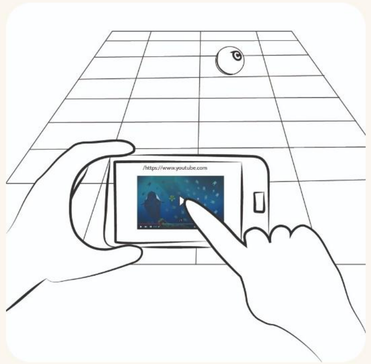
02 Roll and drive

The die gives a number; the child steers Sphero that many blocks.



03 Scan the block

Every block hides a QR code under a character or object.



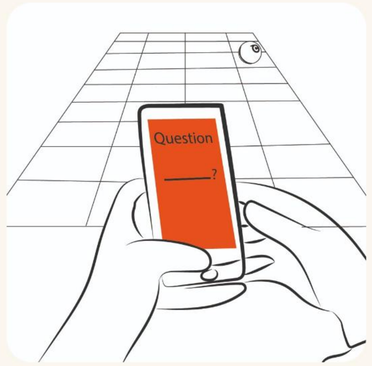
04 Watch

Some blocks open a short film about the ocean.



05 Read

Others open a story or a short paragraph to read together.



06 Answer

Questions check what was learned before the next roll.

EACH STAGE 1 2 3 4 5 6 7 8 ★

A run of activity blocks, closed by a badge — then on to the next, deeper stage.

1 m

4 m



STAGE 1 · SHORE

Lesson 1 — Familiarity with the ocean

Umbrella, deckchair, crabs and the talking shell: why the ocean matters to everyone.

STAGE 2 · SHALLOWS

Lesson 2 — Ocean animals

A snorkeler's view: walrus, turtle, coral, a cave — life just below the surface.

STAGE 3 · OPEN SEA

Lesson 3 — Types of garbage

Divers and jellyfish — and the garbage monster rises out of the blue.

STAGE 4 · THE DEEP

Lesson 4 — Recycling

Whale, octopus, anglerfish, and a treasure chest waiting at the end.

06 — THE MAP

Four metres, shore to deep

The map is the game board and the story at once. It begins on familiar ground — a sunny beach — and sinks, stage by stage, into water children have never seen.

Its colour darkens with depth, so a child always knows how far they have come and how far is left.

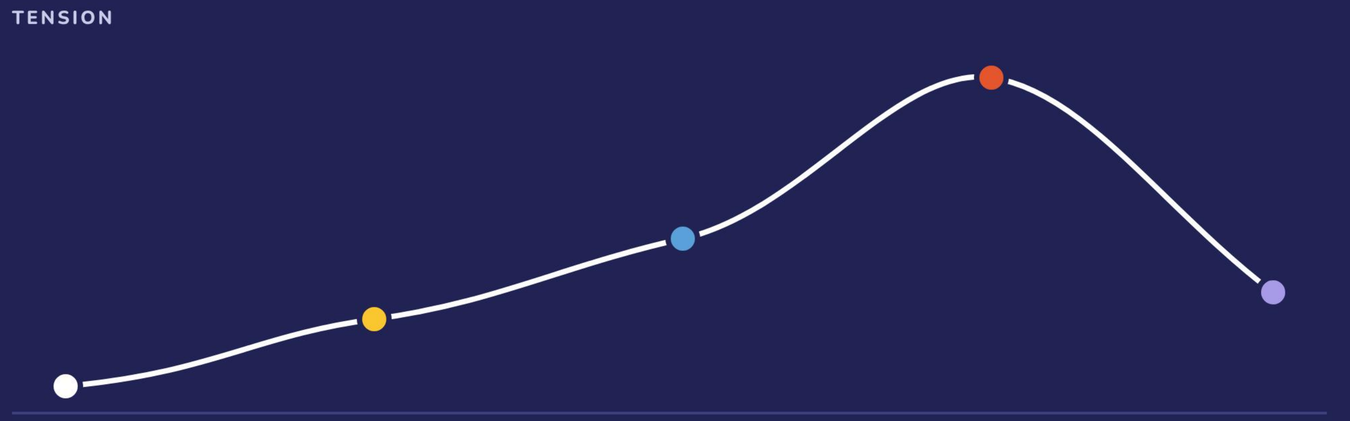
4	4 × 1 m	1
connected stages	printed size	continuous journey

07 — THE STORY

A shell asks for help

> “Hi, dear friend — I'm a shell! Sadly, my friends and I are not in a good situation. Every day we're more surrounded by the big black monster of garbage. Can you help us?”

The whole programme hangs on this one message. Children don't study the ocean; they answer a friend's call, and every lesson becomes a step in a rescue.



THE HERO

The child — an ocean defender on a mission



THE GUIDE

The shell, who speaks through a QR code

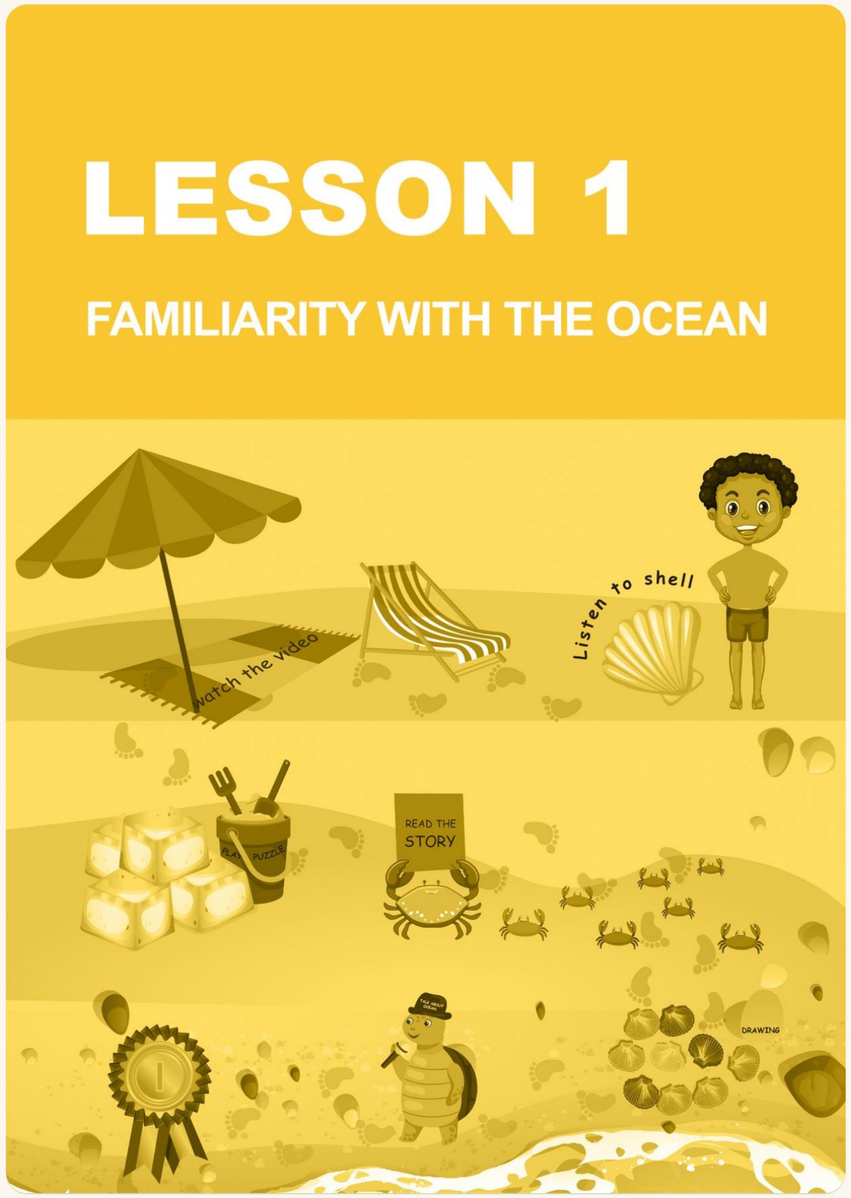
THE HELPER

The teacher, who reads the guidebook aloud



THE ANTI-HERO

A giant black monster made of garbage







Lesson board · activity sheet · the stage in full colour

08 — STAGE 1 · SHORE

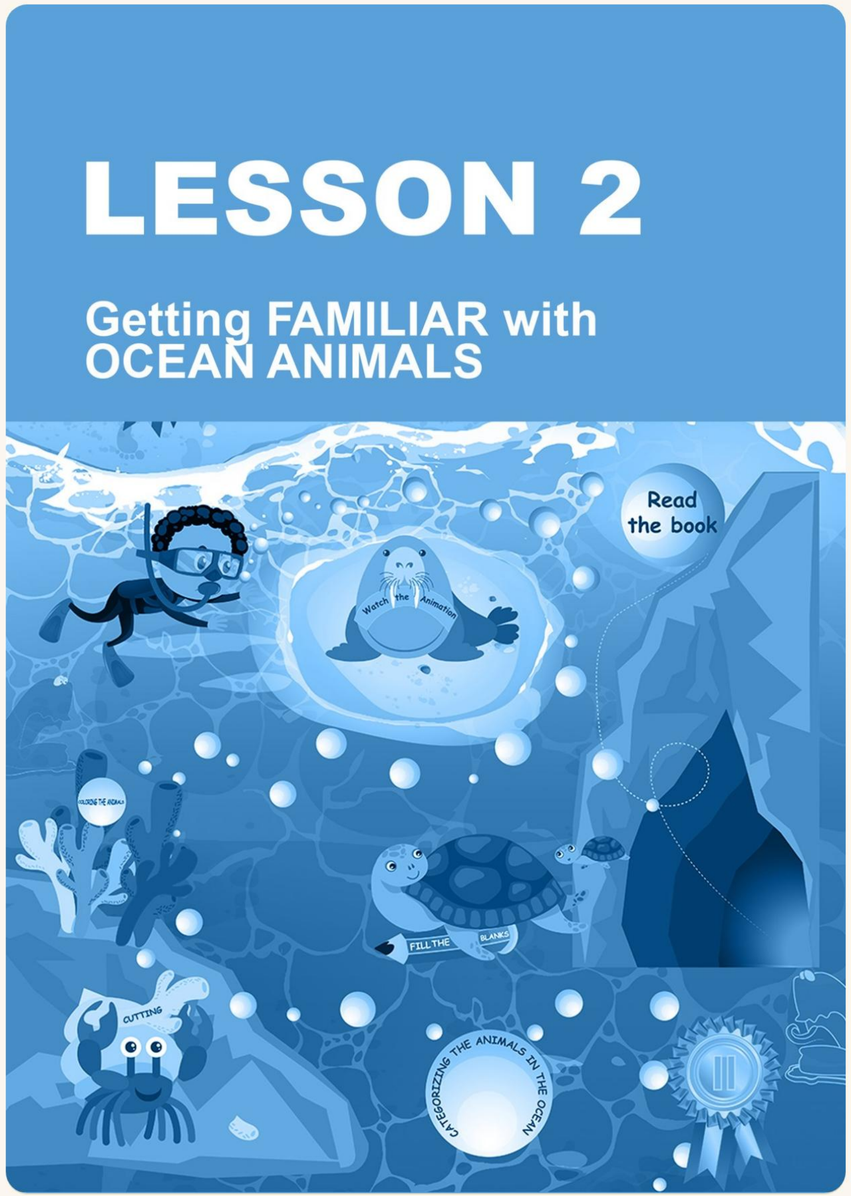
Familiarity with the ocean

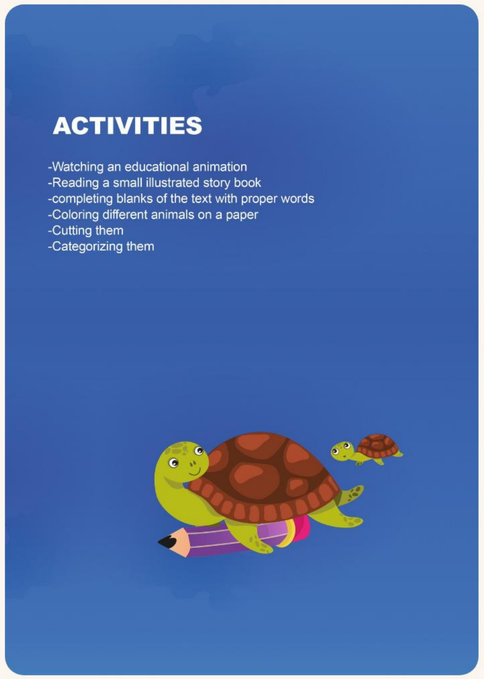
Warm yellow, open sand, familiar things. The journey starts where children already feel at home — the beach — and explains why the ocean matters to all of us.

THE CHILD'S ROLE A beach-goer: children come to class in their swimsuits.

BLOCKS AND PACING

1	The shell's message	5 min
2	Video under the umbrella	5 min
3	Video, continued	5 min
4	Nine puzzle cubes	3 min
5	The crab's story card	5 min
6	Story, continued	6 min
7	Drawing the ocean	15 min
8	A talk to friends	8 min
9	**Badge — stage complete**	**52 min**







Lesson board · activity sheet · the stage in full colour

09 — STAGE 2 · SHALLOWS

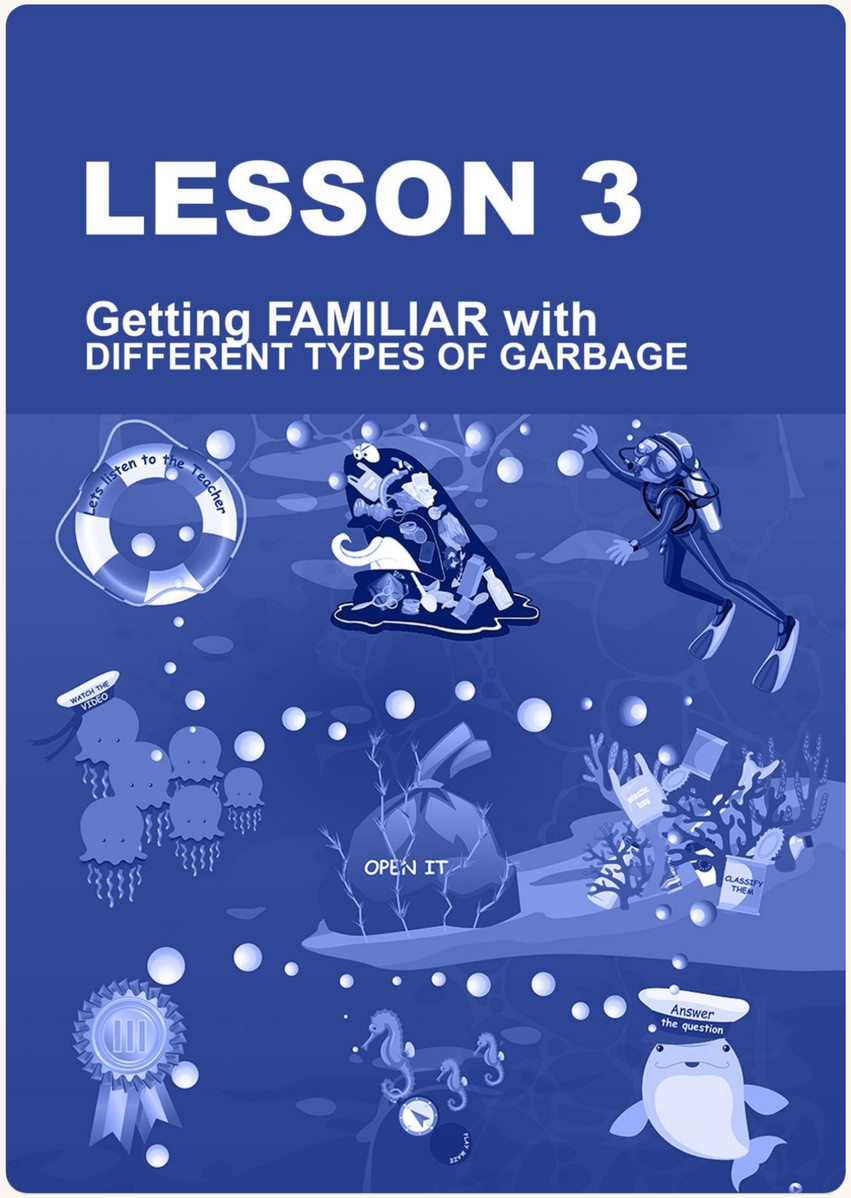
Ocean animals

The water turns light blue and animals take over the map. Children learn how much lives below the surface — and how little of it we know.

THE CHILD'S ROLE A snorkeler swimming in the first metres of the ocean.

BLOCKS AND PACING

11	Snorkel glasses: animation	7 min
12	Illustrated story book	6 min
13	Story book, continued	6 min
14	Fill in the blanks	5 min
15	Colour ten sea animals	10 min
16	Cut them out	5 min
17	Sort them by depth	11 min
18	**Badge — stage complete**	**50 min**







Lesson board · activity sheet · the stage in full colour

10 — STAGE 3 · OPEN SEA

Types of garbage

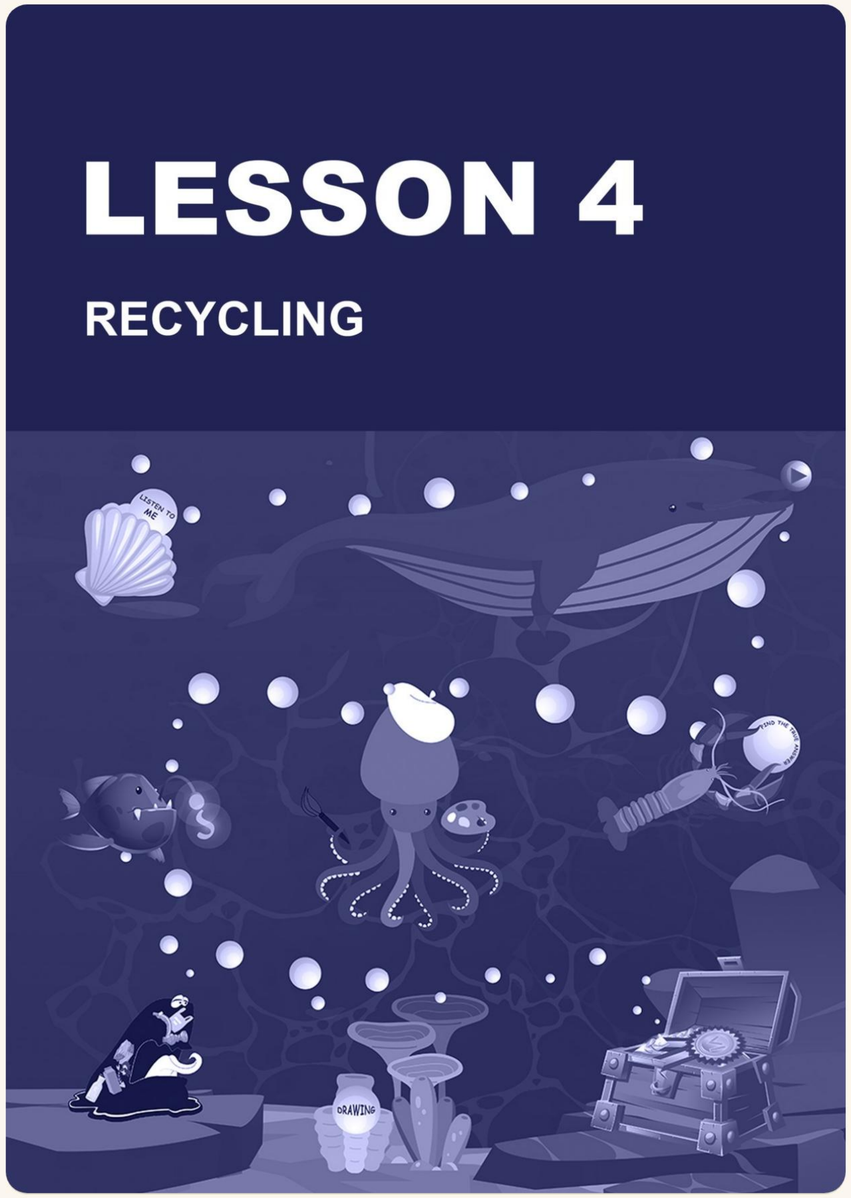
The blue deepens and the story turns. Pollution becomes visible on the map, and children learn to tell one kind of garbage from another.

THE TURN IN THE STORY

The anti-hero arrives: a giant black monster made of garbage.

ACTIVITIES

1. Watch a video on kinds of garbage and how long they last underwater
2. Hear from the teacher about each type
3. Sort all the garbage by material
4. Answer ten questions
5. Solve a maze game







Lesson board · activity sheet · the stage in full colour

11 — STAGE 4 · THE DEEP

Recycling

Deep blue turning violet. Having seen the problem, children learn what to do about it — separate, recycle, and rescue the shell's friends.

THE ENDING The shell comes back — this time happy and laughing.

ACTIVITIES

1. Listen to the shell on why recycling matters
2. Watch a video on recycling and its levels
3. Solve a puzzle
4. Draw
5. Answer ten questions
6. Solve a final puzzle

12 — CHARACTERS

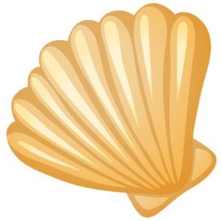
A cast that shows the way

Every creature is drawn the same way: flat shapes, soft shading, round friendly eyes — friendly enough to approach, simple enough to read from across a classroom floor.

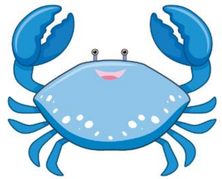
Many carry the activity with them: a crab holds up the story card, the octopus brings a brush and palette, the seahorse a compass. A child can read the next task from the picture before reading a single word.



Hermit crab



The shell



Blue crab



Deckchair



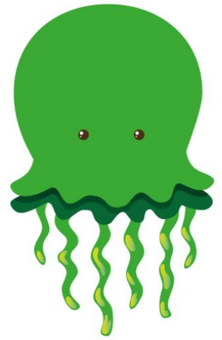
Coral



Life ring



Octopus artist



Jellyfish



Seahorse & compass



Anglerfish

13 — COLOUR & WAYFINDING

Colour as a depth gauge

Four colours carry the whole journey: warm yellow on the beach, light blue in the shallows, deep blue in open water and violet-navy at the bottom. Each lesson board is tinted in its stage's colour, so the boards alone tell a child which part of the sea they are in.

STAGE 1

Shore

#FAC730

STAGE 2

Shallows

#5AA1D9

STAGE 3

Open sea

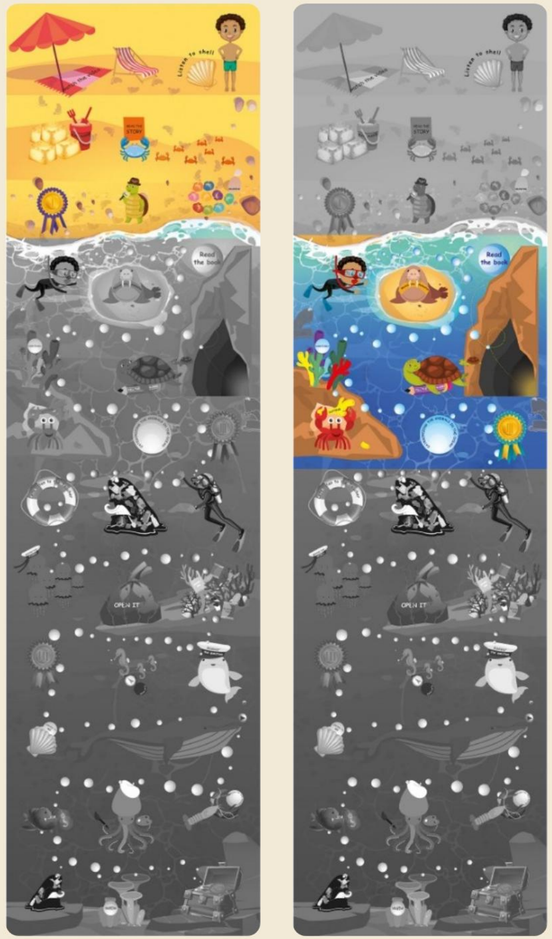
#304797

STAGE 4

The deep

#212354





YOU ARE HERE

One stage in colour, the rest in grey

In the lesson materials the map is shown whole, with only the current stage in colour. Children see the entire journey and exactly where they stand in it.

14 — PROCESS

Eight months, eight steps

2020–2021. Each step fed the next: the story decided what the map needed to show, and the map decided what had to be drawn.

1

01 Research & scenario

Oceans, pollution and sustainability; what the programme had to teach.

2

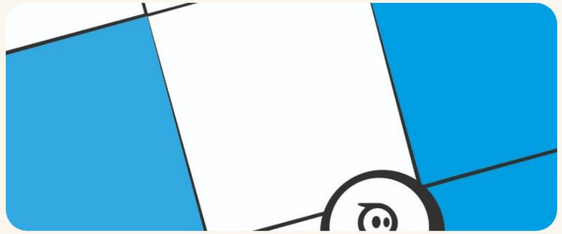
02 Audience & goals

Nine-year-olds; four learning objectives to build around.



03 Story

A guide, a hero, a helper and an anti-hero to carry the lessons.



04 Storyboard

Line sketches of every interaction: roll, drive, scan, watch, read, answer.



05 Map structure

Four stages, each a run of activity blocks ending in a badge.



06 Illustration

The marine cast, props and four environments in one style.



07 Lesson boards & sheets

A tinted board and an activity sheet for every lesson.

8

08 Lesson plan

Minutes for every block, so a teacher can run a stage in one session.

15 — REFLECTION

What this project shows



Visual storytelling

A programme of facts turned into a rescue story with a guide, a hero and an anti-hero.

Illustration system

One cast and four environments, consistent across a four-metre map, lesson boards and sheets.

Learning experience

A play loop and pacing built for nine-year-olds: physical, short, rewarded at every stage.

Systems thinking

Map, robot, QR codes, guidebook and worksheets designed to work as one game.

WHERE IT COULD GO NEXT

- → Playtest a full stage with a class and time it against the plan
- → Check the four stage colours for colour-blind readers
- → Make the shell's voice and the videos available offline